Rain- Based Traffic Alert System

Team No: 6

Team Name: Nova

Team Members:

1. Manisha N - 2360412
2. Indla Sai Sneha - 2362080
3. Reyona - 2360601
4. Libena Igi J – 2362519

MileStone 4: Analytics Output and Dashboard Visualization

1. Introduction

The fourth milestone of the Rain-Based Traffic Alert System focuses on transforming the generated dataset and machine learning outputs into meaningful visual insights using Power BI. While the previous phase concentrated on dataset generation and predictive modeling, this stage emphasizes data interpretation, monitoring, and decision support.Data visualization plays a critical role in intelligent systems by converting raw numerical data into graphical representations that are easy to understand. In this project, Power BI is used to create an interactive dashboard that provides real-time insights into traffic risk levels based on environmental and vehicular parameters such as rainfall, speed, distance, and temperature.This milestone bridges the gap between **data analysis and real-world application**, enabling authorities to take immediate actions based on visual insights.

2. Objectives of the dashboard

The main objectives of this visualization system are:

- To convert complex IoT and ML-generated data into **interactive visual insights**
- To identify **high-risk traffic zones** across different locations
- To analyze the **impact of rainfall on road safety**
- To provide a **real-time alert mechanism**
- To assist traffic authorities in **efficient decision-making and response planning**

- To enable **dynamic filtering and exploration** of data

3. Dataset Overview

As developed in Milestone 2 , the dataset contains multiple features collected from simulated IoT sensors.

Key Features Visualized:

- Rain (converted to mm and cm)
- Traffic (vehicle presence)
- Distance between vehicles
- Temperature
- Humidity
- Visibility
- Speed
- Risk Level (Safe, Caution, High, Critical)
- Location (Bangalore regions)
- Patrol Unit
- Phone Number

The dataset consists of **1500 rows**, covering various real-world traffic scenarios.

4. Dashboard Architecture

The Power BI dashboard is designed using multiple visual components that collectively provide a complete understanding of the system.

4.1 Data Flow Architecture

1. IoT Sensors (Simulated Data)
2. Dataset Generation (Python)
3. Machine Learning Model (Risk Prediction)
4. Excel/CSV Storage
5. Power BI Dashboard (Visualization Layer)

This architecture ensures smooth data transformation from raw input to actionable insights.

5. Detailed Visualization Components

5.1 KPI Summary Section

The dashboard begins with key aggregated metrics:

- Sum of Rain Percentage
- Sum of Speed
- Sum of Temperature

These KPIs provide a quick snapshot of environmental and traffic conditions.

Interpretation:

- Higher rainfall and speed combinations indicate unsafe driving conditions
- Temperature variations may affect road conditions and visibility

5.2 Rainfall Analysis by Location

Visualization Type: Bar Chart

Field Used:

- X-axis → Location
- Y-axis → Sum of Rain (mm)

Purpose:

To compare rainfall intensity across different areas.

Insights:

- Locations like **MG Road and Whitefield** experience higher rainfall levels
- Areas with higher rainfall are more prone to accidents
- Helps authorities prioritize high rainfall zones

5.3 Risk Distribution Analysis

Visualization Type: Donut Chart

Field Used:

- Legend → Risk
- Values → Count of Patrol Units

Purpose:

To analyze how risk levels are distributed across the dataset.

Insights:

- A larger proportion of data falls under **High Risk**
- Safe conditions are comparatively fewer
- Indicates that the system effectively captures critical scenarios

5.4 Rain vs Distance vs Risk Relationship

Visualization Type: Scatter Plot

Fields Used:

- X-axis → Distance
- Y-axis → Rain (mm)
- Size/Color → Risk Count

Purpose:

To understand the relationship between rainfall and vehicle distance.

Insights:

- Lower distance between vehicles increases collision risk
- High rainfall combined with short distance leads to **High/Critical risk**
- Validates the logic defined in the ML model

5.5 Geographic Visualization (Map)

Visualization Type: Map (Bing Map)

Purpose:

To display traffic conditions geographically across Bangalore.

Insights:

- Identifies **risk hotspots**
- Helps in route planning and traffic diversion
- Useful for real-time monitoring by authorities

5.6 Interactive Filters (Slicers)

Filters Used:

- Risk Level (High, Caution, Safe)
- Location

Purpose:

To allow users to customize the dashboard view.

Benefits:

- Focus on specific regions
- Analyze particular risk levels
- Improves user interaction and usability

5.7 Traffic Indicator (Gauge Chart)

Visualization Type: Gauge Chart

Field Used: Sum of Traffic

Purpose:

To represent current traffic intensity.

Insights:

- Indicates congestion level
- Helps understand road load conditions
- Useful for traffic control planning

5.8 Smart Alert System

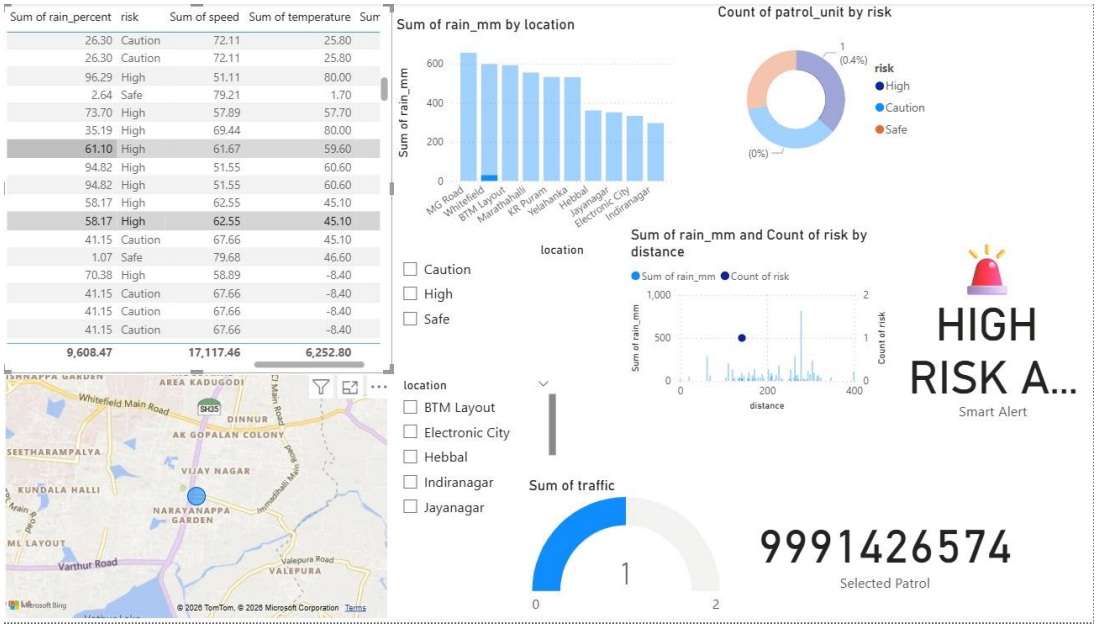
The dashboard includes a **dynamic alert display**:

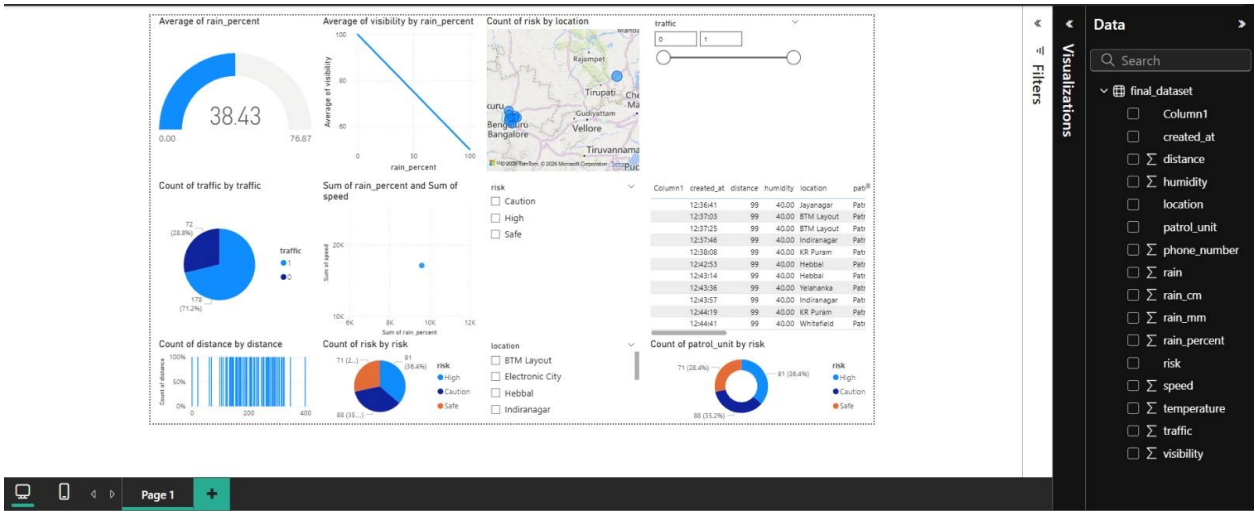
- Displays: **"HIGH RISK ALERT"**
- Shows **Patrol Unit Phone Number**

Purpose:

- Immediate notification system
- Enables quick response by authorities
- Supports accident prevention

This directly aligns with the alert mechanism described in Milestone 2.





6. Correlation with Machine Learning Model

The visualization supports and validates the ML model outcomes:

- High rainfall + low distance → High risk
- Moderate conditions → Caution
- Normal conditions → Safe

The dashboard visually confirms these patterns, ensuring the correctness of the model predictions.

7. Key Observations and Insights

- Rainfall is the most influential factor in traffic risk prediction
- Short distance between vehicles significantly increases accident probability
- Certain locations consistently show higher risk levels
- High-risk scenarios dominate the dataset, indicating real-world applicability

- Patrol units can be efficiently deployed based on risk concentration

8. Advantages of the Dashboard

- Provides **real-time data interpretation**
- Enhances **decision-making capability**
- Easy to understand even for non-technical users
- Interactive and dynamic
- Integrates multiple data sources into a single view
- Supports **smart city and intelligent transport systems**

9. Limitations

- Dataset is synthetically generated (not real-time)
- Map accuracy depends on input data
- Alerts are not yet automated in real-world systems
- Does not include live IoT integration

10. Future Enhancements

- Integration with **real-time IoT sensors**
- Live data streaming into Power BI
- AI-based predictive alerts
- Mobile app integration for users
- Automated emergency response system
- Cloud deployment using Azure

11. Conclusion

The Power BI dashboard successfully enhances the Rain-Based Traffic Alert System by converting machine learning outputs into actionable insights. It provides a comprehensive view of environmental conditions, traffic behavior, and risk distribution.

By integrating visualization with IoT and machine learning, the system demonstrates a practical approach toward improving road safety. The dashboard not only helps in identifying dangerous conditions but also enables quick response through alert mechanisms.

This milestone highlights the importance of visualization in transforming data into intelligent decision-making tools, making it a crucial component of modern smart traffic system